

Descriptive Statistics

Monday 27th October, 2014

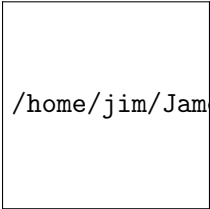
Section 4.1: Population, Sample, and Data

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Monday 27th October, 2014



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Descriptive Statistics:

Descriptive Statistics:

Inferential Statistics

Descriptive Statistics:

- collecting,
- organizing,
- and summarizing data

Inferential Statistics

Descriptive Statistics:

- collecting,
- organizing,
- and summarizing data

Inferential Statistics

- drawing conclusions
- and
- making predictions

Population vs Sample

Example

A CBS poll states that 44% of NC voters will vote for Hagan, 41% for Tillis, and 2% for Haugh.

Population vs Sample

Example

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Descriptive Statistics:

How was the data collected?

Population vs Sample

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Descriptive Statistics:

How was the data collected?

Inferential Statistics

Population vs Sample

Example

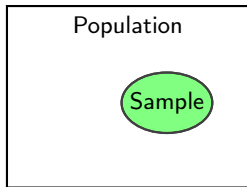
A CBS poll states that 44% of NC voters will vote for Hagan, 41% for Tillis, and 2% for Haugh.

Descriptive Statistics:

How was the data collected?

How can it be organized and displayed?

Inferential Statistics



Population vs Sample

Example

A CBS poll states that 44% of NC voters will vote for Hagan, 41% for Tillis, and 2% for Haugh.

Descriptive Statistics:

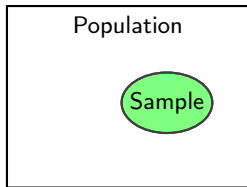
How was the data collected?

How can it be organized and displayed?

Inferential Statistics

How accurate is the poll?

How predictive is it?



Population vs Sample

Population vs Sample

Population

The set of all objects under study.

Population vs Sample

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The set of all objects under study.

Sample

Any subset of the population.

Step 1 of any study is data collection.

Population vs Sample

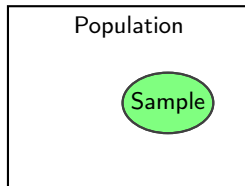
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Any subset of the population.

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Population vs Sample

Population

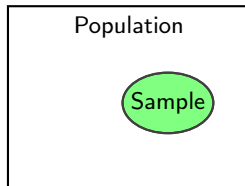
The set of all objects under study.

Sample

Any subset of the population.

- We want a sample that is *representative* of the population.

Step 1 of any study is data collection.



Population vs Sample

Population

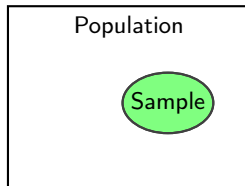
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Any subset of the population.

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- Otherwise, any conclusions drawn will be useless.

Step 1 of any study is data collection.



Population vs Sample

Population

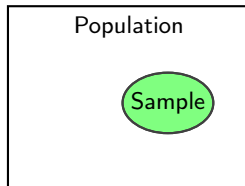
The set of all objects under study.

Sample

Any subset of the population.

- We want a sample that is *representative* of the population.
- Otherwise, any conclusions drawn will be useless.

Step 1 of any study is data collection.



Step 2 is to organize it.

Data Organization

Data Organization

Example 4.1.1.

Number of children under five of each employee at a factory:

0	2	1	0	3
0	1	1	2	4
1	0	1	1	0
2	1	0	0	3
3	0	0	1	2

Example 4.1.1.

Number of children under five of each employee at a factory:

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Data Organization

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0	2	1	0	3
0	1	1	2	4
1	0	1	1	0
2	1	0	0	3
3	0	0	1	2

Each element in this set is called a **data point**.

All the data is the **data set**.

A table that lists each data point and its frequency is called a **frequency distribution**.

Num	Frequency	Relative Freq.
0	9	
1		
2		
3		
4		

Data Organization

Example 4.1.1.

Number of children under five of each employee at a factory:

0	2	1	0	3
0	1	1	2	4
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Num	Frequency	Relative Freq.
0	9	
1	8	
2	4	
3	3	
4	1	

Data Organization

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Num	Frequency	Relative Freq.
0	9	
1	8	
2	4	
3	3	
4	1	
$n = 25$		

Data Organization

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Number of children under five of each employee at a factory:

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1	0	1	1	0
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3	0	0	1	2

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Num	Frequency	Relative Freq.
0	9	$9/25 = 36\%$
1	8	
2	4	
3	3	
4	1	
$n = 25$		

Data Organization

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Number of children under five of each employee at a factory:

0	2	1	0	3
0	1	1	2	4
1	0	1	1	0
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Num	Frequency	Relative Freq.
0	9	$9/25 = 36\%$
1	8	$8/25 = 32\%$
2	4	$4/25 = 16\%$
3	3	$3/25 = 12\%$
4	1	$1/25 = 4\%$
$n = 25$		

Data Organization

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1	8	$8/25 = 32\%$
2	4	$4/25 = 16\%$
3	3	$3/25 = 12\%$
4	1	$1/25 = 4\%$
$n = 25$		total = 100%

Grouped Data Organization

Grouped Data Organization

Example 4.1.2.

Grades from an exam are as follows:

89	85	33	51	74
40	92	64	68	92
84	96	98	60	85
71	96	24	87	81
64	71	58	40	

Grouped Data Organization

Example 4.1.2.

Grades from an exam are as follows:

89	85	33	51	74
40	92	64	68	92
84	96	98	60	85
71	96	24	87	81
64	71	58	40	

Max= 98 and

Min= 24

Range: 98 – 24 = 74

- Grouping data is necessary for more varied data sets;
- usually with 4 to 8 groups.

Grade	Freq.	Relative Freq.

Grouped Data Organization

Example 4.1.2.

Grades from an exam are as follows:

89	85	33	51	74
40	92	64	68	92
84	96	98	60	85
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64	71	58	40	

Max= 98 and

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Range: $98 - 24 = 74$

$74 \div 5 = 14.8 \approx 15$

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Grade	Freq.	Relative Freq.
$24 \leq x < 39$		

Grouped Data Organization

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Grade	Freq.	Relative Freq.
$24 \leq x < 39$		
$39 \leq x < 54$		
$54 \leq x < 69$		
$69 \leq x < 84$		
$84 \leq x < 99$		

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Grade	Freq.	Relative Freq.
$24 \leq x < 39$	2	
$39 \leq x < 54$	3	
$54 \leq x < 69$	5	
$69 \leq x < 84$	4	
$84 \leq x < 99$	10	

Grouped Data Organization

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$24 \leq x < 39$	2	
$39 \leq x < 54$	3	
$54 \leq x < 69$	5	
$69 \leq x < 84$	4	
$84 \leq x < 99$	10	
	$n = 24$	

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Max= 98 and

Min= 24

Range: $98 - 24 = 74$

$74 \div 5 = 14.8 \approx 15$

- Grouping data is necessary for more varied data sets;
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Grade	Freq.	Relative Freq.
$24 \leq x < 39$	2	$2/24 = 8.3\%$
$39 \leq x < 54$	3	$3/24 = 12.5\%$
$54 \leq x < 69$	5	$5/24 = 20.8\%$
$69 \leq x < 84$	4	$4/24 = 16.7\%$
$84 \leq x < 99$	10	$10/24 = 41.7\%$
	$n = 24$	

Grouped Data Organization

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40	92	64	68	92
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$24 \leq x < 39$	2	$2/24 = 8.3\%$
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$54 \leq x < 69$	5	$5/24 = 20.8\%$
$69 \leq x < 84$	4	$4/24 = 16.7\%$
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Grade	Freq.	Relative Freq.
24 – 38.9	2	$2/24 = 8.3\%$

Grouped Data Organization

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40	92	64	68	92
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Max= 98 and

Min= 24

Range: $98 - 24 = 74$

$74 \div 5 = 14.8 \approx 15$

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Grade	Freq.	Relative Freq.
24 – 38.9	2	$2/24 = 8.3\%$
39 – 53.9	3	$3/24 = 12.5\%$
54 – 68.9	5	$5/24 = 20.8\%$
69 – 83.9	4	$4/24 = 16.7\%$
84 – 98.9	10	$10/24 = 41.7\%$

Grouped Data Organization

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Grades from an exam are as follows:

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40	92	64	68	92
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64	71	58	40	

Max= 98 and

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Range: 98 – 24 = 74

$74 \div 5 = 14.8 \approx 15$

- Grouping data is necessary for more varied data sets;
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Grade	Freq.	Relative Freq.
24 – 38.9	2	$2/24 = 8.3\%$
39 – 53.9	3	$3/24 = 12.5\%$
54 – 68.9	5	$5/24 = 20.8\%$
69 – 83.9	4	$4/24 = 16.7\%$
84 – 98.9	10	$10/24 = 41.7\%$
	$n = 24$	total = 100%

Grouped Data Organization

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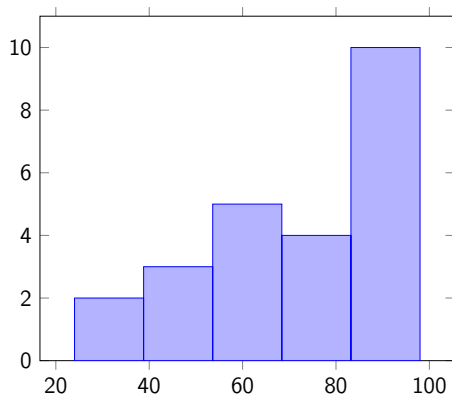
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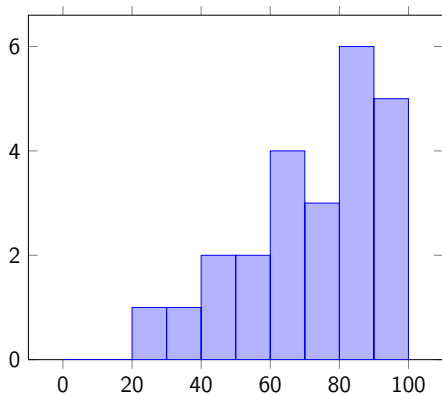
Grade	Freq.	Relative Freq.
0 – 69.9	10	$10/24 = 41.7\%$
70 – 79.9	3	$3/24 = 12.5\%$
80 – 89.9	6	$6/24 = 25\%$
90 – 100	5	$5/24 = 20.8\%$
$n = 24$		total = 100%

Histograms

Histograms



Histograms



Grade	Freq.	Rel.
0 – 69.9	10	41.7%
70 – 79.9	3	12.5%
80 – 89.9	6	25%
90 – 100	5	20.8%

Example 4.1.3.

200 randomly selected MHU graduates are listed below. Where possible, determine what percent of the graduates had the following ages:

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Example 4.1.3.

200 randomly selected MHU graduates are listed below. Where possible, determine what percent of the graduates had the following ages:

- a. Less than 23

$$\frac{1+4+52}{200} = 0.285 = 28.5$$

Example 4.1.3.

200 randomly selected MHU graduates are listed below. Where possible, determine what percent of the graduates had the following ages:

- a. Less than 23

$$\frac{1+4+52}{200} = 0.285 = 28.5$$

- b. at most 20

$$\frac{?}{200}, \text{ not possible}$$

Example 4.1.3.

200 randomly selected MHU graduates are listed below. Where possible, determine what percent of the graduates had the following ages:

- a. Less than 23

$$\frac{1+4+52}{200} = 0.285 = 28.5$$

- b. at most 20

$$\frac{?}{200}, \text{ not possible}$$

- c. at least 19 but less than 27

$$\frac{52+48}{200} = 0.5 = 50\%$$

Histograms

Example 4.1.3.

200 randomly selected MHU graduates are listed below. Where possible, determine what percent of the graduates had the following ages:

- a. Less than 23

$$\frac{1+4+52}{200} = 0.285 = 28.5$$

- b. at most 20

$$\frac{?}{200}, \text{ not possible}$$

- c. at least 19 but less than 27

$$\frac{52+48}{200} = 0.5 = 50\%$$

Grade	Freq.
$10 \leq x < 15$	1
$15 \leq x < 19$	4
$19 \leq x < 23$	52
$23 \leq x < 27$	48
$27 \leq x < 31$	31
$31 \leq x < 35$	16
$35 \leq x < 39$	29
39 and over	19